Feature Selection in Statistical Machine Learning

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Lecture 20

Recap and outline

- Exam outline
- Shapley values for feature attribution
- Shapley values and feature selection
- Examples

Exam overview

- Logistics
- What you can and can't bring
- Academic misconduct policy
- General tips

Exam logistics and format

- Wednesday April 29 8:05-9:05 (60 minutes) in usual classroom
- 3 questions worth 20 points each
- See practice final/homework 4

Exam what you can and can't bring

- No technology (including calculators) allowed
- You can bring one double-sided US letter cheat sheet
- You will be provided with a notebook to provide answers in

Exam academic misconduct policy

- No technology is allowed and AI use is strictly forbidden
- If you are found to use AI either during or after the exam, you will receive a 0 for the exam and be reported for academic misconduct
- Trust yourself that things will be ok and if the problem seems difficult, others may also find it difficult
- Same applies to collaborating and looking at other people's work

Exam general tips

- Practice final/homework 4 is most closely aligned with final exam
- Examples from lecture and homework questions that do not involve code are also useful resources
- All material and ideas is self-contained within lectures and homeworks

Shapley value axioms

- General framework for feature attribution measures that satisfy 4 properties/axioms
  1. Efficiency
  $$\sum_{j=1}^{p} \phi_j = \widehat{f}(\tilde{X}_1, \tilde{X}_2, ..., \tilde{X}_p)$$
  2. Symmetry
  If X_j and X_k are interchangeable then
  $$\phi_j = \phi_k$$
  3. Dummy
  If X_k contains no information about Y
  $$\phi_k = 0$$
  4. Additivity
  If $f = f_1 + f_2$, for all j
  $$\phi_j(f) = \phi_j(f_1) + \phi_j(f_2)$$

Shapley value general formula

Feature set $(X_1, X_2, ..., X_p)$
$S \subset \{1, 2, ..., p\}$ denotes any subset of indices
Value function:

$$v : 2^p \to \mathbb{R}$$

is a function over a subset of the features (will see various examples later)
Shapley value:

$$\phi_j(v) = \sum_{S \subset \{1,2,..,p\}/\{j\}} \frac{|S|!(p - |S| - 1)!}{p!}(v(S \cup \{j\}) - v(S))$$

Provably satisfies all 4 properties

High-level intuition

Shapley value:

$$\phi_j(v) = \sum_{S \subset \{1,2,..,p\}/\{j\}} \frac{|S|!(p-|S|-1)!}{p!}(v(S \cup \{j\}) - v(S))$$

- $(v(S \cup \{j\}) - v(S))$ is a of how much feature X_j improves utility when working with the features X_S
- $\frac{|S|!(p-|S|-1)!}{p!} = \frac{1}{p}\binom{p-1}{|S|}$ is a weight that takes accounts for the size of S relative to $p-1$

Example choices of $v(.)$ functional

Conditional value:

$$v(S) = \mathbb{E}[f(X)|X_S = x_S]$$

Global risk (naturally connects to feature selection):

$$v(S) = -\mathbb{E}[\ell(f_S(X_S), Y)]$$

Functional:

$$v(S) = f(x_S, x'_{S^c})$$

where x' is baseline/local point of interest
Functional (connected to integrated gradient):

$$v(S) = f(x' + 1_S \odot (x - x'))$$

where x' is baseline/local point of interest
In this course, we will mainly focus on first 2

Example 1: 2-feature linear model

$$Y = \beta_1 X_1 + \beta_2 X_2$$

where $\mathbb{E}[X_1] = \mathbb{E}[X_2] = 0$

$$v(S) = \mathbb{E}[f(X)|X_S = x_S]$$

$$\begin{aligned}
v(\emptyset) &= \mathbb{E}[\beta_1 X_1 + \beta_2 X_2] = 0 \\
v(\{1\}) &= \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_1 = x_1] = \beta_1 x_1 + \beta_2 \mathbb{E}[X_2 | X_1 = x_1] \\
v(\{2\}) &= \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_2 = x_2] = \beta_1 \mathbb{E}[X_1 | X_2 = x_2] + \beta_2 x_2 \\
v(\{1,2\}) &= \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_1 = x_1, X_2 = x_2] = \beta_1 x_1 + \beta_2 x_2
\end{aligned}$$

Example 1: Independent features

$$
\begin{aligned}
v(\emptyset) &= \mathbb{E}[\beta_1 X_1 + \beta_2 X_2] = 0 \\
v(\{1\}) &= \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_1 = x_1] = \beta_1 x_1 + \beta_2 \mathbb{E}[X_2 | X_1 = x_1] = \beta_1 x_1 \\
v(\{2\}) &= \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_2 = x_2] = \beta_1 \mathbb{E}[X_1 | X_2 = x_2] + \beta_2 x_2 = \beta_2 x_2 \\
v(\{1,2\}) &= \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_1 = x_1, X_2 = x_2] = \beta_1 x_1 + \beta_2 x_2
\end{aligned}
$$

Following Shapley value calculation:

$$\phi_1 = \beta_1 x_1$$

and

$$\phi_2 = \beta_2 x_2$$

Example 1: Perfectly correlated features $X_1 = X_2$

$$
\begin{aligned}
v(\emptyset) &= \mathbb{E}[\beta_1 X_1 + \beta_2 X_2] = 0 \\
v(\{1\}) &= \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_1 = x_1] = \beta_1 x_1 + \beta_2 \mathbb{E}[X_2 | X_1 = x_1] = (\beta_1 + \beta_2) x_1 \\
v(\{2\}) &= \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_2 = x_2] = \beta_1 \mathbb{E}[X_1 | X_2 = x_2] + \beta_2 x_2 = (\beta_1 + \beta_2) x_2 \\
v(\{1,2\}) &= \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_1 = x_1, X_2 = x_2] = \beta_1 x_1 + \beta_2 x_2
\end{aligned}
$$

Following Shapley value calculation:

$$\phi_1 = \frac{1}{2}(\beta_1 + \beta_2) x_1$$

and

$$\phi_2 = \frac{1}{2}(\beta_1 + \beta_2) x_1$$

Equal credit assignment principle

- In independent case, we get $\phi_j = \beta_j x_j$ as expected (what we used in previous lecture)
- In perfectly correlated case $X_1 = X_2$, $\phi_1 = \phi_2$
- Follows from symmetry property
- Equal credit assignment principle of Shapley values
- **Question:** How do feature selection methods we have seen so far handle perfectly correlated features (e.g. LOCO, Lasso, e.t.c.)?

Example 2: Pairwise interaction model

Interactions occur in practice. Recall negation example for language.
Consider model:

$$Y = X_1 \oplus X_2$$

where X_1, X_2 are independent coin flips.

- $\oplus$ is XOR meaning if $X_1 + X_2$, $Y = 0$ and if $X_1 \neq X_2$, $Y = 1$
- **Question:** What is the Shapley value with $v(S) = \mathbb{E}[f(X)|X_S = x_S]$

Example 2 cont.

$$
\begin{aligned}
v(\emptyset) &= \mathbb{E}[X_1 \oplus X_2] = \frac{1}{2} \\
v(\{1\}) &= \mathbb{E}[X_1 \oplus X_2 | X_1 = x_1] = \frac{1}{2} \\
v(\{2\}) &= \mathbb{E}[X_1 \oplus X_2 | X_2 = x_2] = \frac{1}{2} \\
v(\{1,2\}) &= \mathbb{E}[X_1 \oplus X_2 | X_1 = x_1, X_2 = x_2] = x_1 \oplus x_2
\end{aligned}
$$

Following Shapley value calculation:

$$\phi_1 = \frac{1}{2}(x_1 \oplus x_2 - \frac{1}{2})$$

and

$$\phi_2 = \frac{1}{2}(x_1 \oplus x_2 - \frac{1}{2})$$

Example 2 cont.

Shapley values:

$$\phi_1 = \frac{1}{2}(x_1 \oplus x_2 - \frac{1}{2})$$

and

$$\phi_2 = \frac{1}{2}(x_1 \oplus x_2 - \frac{1}{2})$$

- If $x_1 = x_2$, $\phi_1 = \phi_2 = -\frac{1}{4}$
- If $x_1 \neq x_2$, $\phi_1 = \phi_2 = \frac{1}{4}$

Equal credit assignment principle

- Once again notice that $\phi_1 = \phi_2$
- **Question:** How do feature selection methods we have seen so far handle perfectly correlated features (e.g. LOCO, Lasso, e.t.c.)?

Feature selection for interaction model

- Note the conditional independence statements $Y \perp X_1$, $Y \perp X_2$, $Y \perp X_1 | X_2$, $Y \perp X_2 | X_1$, $Y \not\perp \{X_1, X_2\}$
- Feature selection methods would select no features
- One of the weaknesses of feature selection methods is that they don't capture interaction effects well

Summary of results for two models

Recall some of the weaknesses/limitations of feature selection methods:

- Only capture global (not local) impacts
- Highly dependent/correlated features present identifiability challenges
- Interactions (like language model example) not usually captured

Shapley values ”resolve” the issues:

- Local attribution (depends on (x_1, x_2) in all examples)
- For highly correlated features, equal credit assignment splits contributions evenly
- Same for interaction model

But....

- Shapley values address local vs global issue
- Claim that they address dependent features and interactions issue that feature selection methods struggle with (if you search LLMs or speak to most industry people, they would make this claim)
- However Shapley values really just imposing a different assumption (equal credit assignment) to feature selection methods
- Feature selection methods apply different principles (e.g. Lasso applies sparsity, LOCO picks neither, e.t.c.)
- In fact these two problems (highly correlated features or interactions) are essentially impossible to resolve using the idea of looking at influence of single features

To truly resolve these issues...

- Need to start looking at subsets of features
- For both examples (highly correlated features and interaction models) $Y \perp X_1|X_2$ and $Y \perp X_2|X_1$ but $Y \not\perp \{X_1, X_2\}$
- So it is clear there is a relationship when considering the pair $\{X_1, X_2\}$
- To completely resolve the issue need to consider all subsets of features (computationally expensive)
- Both Shapley values and feature selection have the limitation that they attempt to characterize effect/impact of one feature at a time

Next week

- Connecting feature selection and Shapley values
- Use value function:

$$v(S) = -\mathbb{E}[\ell(f_S(X_S), Y)]$$

- Recall feature selection (feature-wise) criteria:

$$Y \perp X_j | X_{-j} \Leftrightarrow \text{Do not select } X_j$$

- **Question:** For Shapley values is it true that:

$$\phi_j = 0 \Leftrightarrow Y \perp X_j | X_{-j}?$$